Assignment: CI/CD on target

Setup

Debug Probe Serial Number	801000233
Host Docker IP Adress	

- Verify with the debugger the connection and that debugging works as expected:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   MEMORY   XRTOS   SERIAL MONITOR

Waiting for gdb server to start...[2026-04-28T06:53:52.978Z] SERVER CONSOLE DEBUG: onBackendConnect: gdb-server session connected. You can switch to "DEBUG CONS
OLE" to see GDB interactions.
"C:\\Raspy\\pico\\sdk\\JLink/JLinkGDBServerCL" -singlerun -nogui -if swd -port 50000 -swoport 50001 -telnetport 50002 -device RP2040_M0_0 -select usb=801000233
-rtos GDBServer/RTOSPlugin_FreeRTOS.dll
SEGGER J-Link GDB Server V8.10g Command Line Version

JLinkARM.dll V8.10g (DLL compiled Nov 14 2024 08:48:50)

Command line: -singlerun -nogui -if swd -port 50000 -swoport 50001 -telnetport 50002 -device RP2040_M0_0 -select usb=801000233 -rtos GDBServer/RTOSPlugin_FreeRT
OS.dll
-----GDB Server start settings-----
GDBInit file:                  none
GDB Server Listening port:     50000
SWO raw output listening port: 50001
Terminal I/O port:             50002
Accept remote connection:      localhost only
Generate logfile:              off
Verify download:               off
Init regs on start:            off
Silent mode:                   off
Single run mode:               on
Target connection timeout:     0 ms
------J-Link related settings------
J-Link Host interface:         USB
J-Link script:                 none
J-Link settings file:          none
oW_Blinky)  [Debug]  Build  [debug build]  [[Targets In Preset]]  [TSM_PicoW_Blinky]  [[Default]]  [[Default]]  Run CPack  [[Default]]  Run Workflow
```

Encountered Problem: the LED blinks only 3 times. But this isn't a fault, it is because i had PL_CONFIG_USE_GCOV activated, which results in the program stopping the scheduler after 6 commutations:

```
static void AppTask(void *pv) {
  int count = 0;
#if PL_CONFIG_USE_PICO_W && PL_CONFIG_USE_LEDS
  Leds_InitFromTask(); /* needed for the on-board Pico-W LED */
#endif
  for(;;) {
    count++;
    #if PL_CONFIG_USE_LEDS
    Leds_Neg(LEDS_ONBOARD);
    #endif
    vTaskDelay(pdMS_TO_TICKS(1000));
    #if PL_CONFIG_USE_GCOV
    if (count>5) {
      vTaskEndScheduler();
    }
    #endif
  }
}
```

Tunnel Server

```
PS C:\Raspy\pico\TSM_PicoW_Blinky> JLinkRemoteServerCL -UseTunnel -TunnelBySN -select="801000233"
SEGGER J-Link Remote Server V8.10g
Compiled Nov 14 2024 08:53:38

'q' to quit '?' for help

2026-04-28 07:15:12 - Remote Server started
2026-04-28 07:15:12 - Connected to J-Link with S/N 801000233
2026-04-28 07:15:12 - Resolving tunnel server name (jlink-europe.segger.com)...
2026-04-28 07:15:12 - Tunnel server IP: 185.162.249.2
2026-04-28 07:15:13 - Registered at tunnel server
2026-04-28 07:15:13 - Waiting for client connection
```

- Launch in VS Code a debug session to verify that your connection through the tunnel service is working.

```
PS C:\Raspy\pico\TSM_PicoW_Blinky> JLinkRemoteServerCL -UseTunnel -TunnelBySN -select="801000233"
SEGGER J-Link Remote Server V8.10g
Compiled Nov 14 2024 08:53:38

'q' to quit '?' for help

2026-04-28 07:21:40 - Remote Server started
2026-04-28 07:21:40 - Connected to J-Link with S/N 801000233
2026-04-28 07:21:40 - Resolving tunnel server name (jlink-europe.segger.com)...
2026-04-28 07:21:40 - Tunnel server IP: 185.162.249.2
2026-04-28 07:21:40 - Registered at tunnel server
2026-04-28 07:21:40 - Waiting for client connection
2026-04-28 07:21:44 - Client connected
```

Local Docker

The dockerfile does the following:

1. It starts Ubuntu 24.04, installs requirements (build toolchain including git, cmake, gcc, and some tools).
2. Installs the good gcov and gcov and SEGGER
3. Clones the Pico SDK
4. Builds a copy of the project source files into /project/
5. Creates documentation with doxygen

Tests are disabled / commented out.

Building docker image with `docker build -t cicd-image -f .\.docker\Dockerfile .`

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   MEMORY   XRTOS   SERIAL MONITOR

=> [26/27] RUN   cd /project   && cmake --preset Debug   && cmake --build --preset app-debug   && cmake --preset Release   && cmake --build --preset app-release   && cmake --preset Test   && cmake --build -   45.1s
=> [27/27] RUN   cd /project/doxy   && doxygen Doxyfile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                1.0s
=> exporting to image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                195.8s
=> => exporting layers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                156.8s
=> => exporting manifest sha256:681e30cd7d9cb666e30cff0ad077235c14ed7484e11eb7e699028cf565b5b087                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => exporting config sha256:1946c5305cfb35fb982c24cb353bc544b9433532b3be6fc461cddafa9849ff5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => exporting attestation manifest sha256:6079782dc8bb968c64d0050380a6442c75f486847be8191c09f0df7c2407d109                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => exporting manifest list sha256:d40011fbdc4cce194e67b2440461d5d004a0c1e0e6462f369e560a7de5091496                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => naming to docker.io/library/cicd-image:latest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => unpacking to docker.io/library/cicd-image:latest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                38.9s
PS C:\Raspy\pico\TSM_PicoW_Blinky>
```

Encountered Problem: it fails connecting to J-link

```
ERROR: cannot open '/project/build/Test/TSM_PicoW_CI_CD.elf' for reading.
root@b6bf9993f232:/project# JRun --verbose --device RP2040_M0_0 --rtt -if SWD --pc off --sp off --ip
"tunnel:801000233" --args "Led_1" /project/build/Test/TSM_PicoW_Blinky.elf
SEGGER J-Run V8.10g Command Line Version
Open application...OK
Connecting to J-Link...
ERROR: Communication timed out: Requested 2 bytes, received 0 bytes !
EMU_GetFirmwareString: Insufficient data read when trying to read the string length.
Cannot connect to J-Link
root@b6bf9993f232:/project# JRun --verbose --device RP2040_M0_0 --rtt -if SWD --pc off --sp off --ip
"tunnel:801000233" --args "Led_1" /project/build/Test/TSM_PicoW_Blinky.elf
SEGGER J-Run V8.10g Command Line Version
Open application...OK
Connecting to J-Link...
ERROR: Unspecified error: -1
Cannot connect to J-Link.
```

Reason: i closed the JLinkRemoteServer on Windows. I restart it in a separate terminal and it worked:

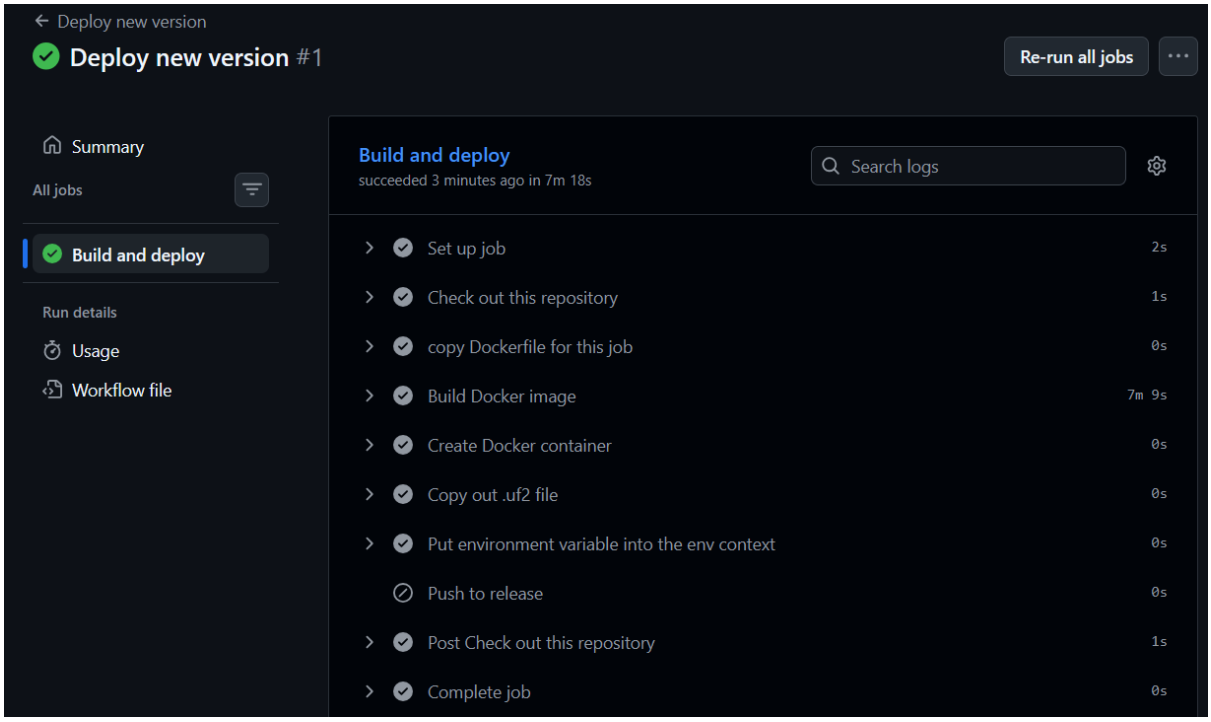
```
root@b6bf9993f232:/project# JRun --verbose --device RP2040_M0_0 --rtt -if SWD --pc off --sp off --ip "tunnel:801000233" --args "Led_1" /pro
ject/build/Test/TSM_PicoW_Blinky.elf
SEGGER J-Run V8.10g Command Line Version
Open application...OK
Connecting to J-Link...OK
Connected to J-Link EDU Mini V1 compiled Oct  9 2024 10:54:30 (S/N 801000233).
JLinkARM.dll V8.10g (DLL compiled Nov 14 2024 08:46:17)
Set target device to RP2040_M0_0...OK
Select SWD interface...OK
Set interface speed to 4000 kHz...OK
Reset target...OK
Downloading file /project/build/Test/TSM_PicoW_Blinky.elf...OK
Set RTT control block at 0x2000214C...OK
Start RTT...OK
Determine application addresses (SP <noinit>, PC <noinit>)...OK
Start target application... OK
Reading output from target until exit command.
=============================================
Target halted on semihosting breakpoint 0x10005CCE.
Operation: SH_SYS_OPEN, R0: 1, R1: 537141172
Target halted on semihosting breakpoint 0x10005CCE.
Operation: SH_SYS_OPEN, R0: 1, R1: 537141172
Target halted on semihosting breakpoint 0x10005CCE.
Operation: SH_SYS_OPEN, R0: 1, R1: 537141172
INFO  tests.c:28 starting test task
*ARGS*Led_1
INFO  McuUnity.c:82 args = Led_1, nofBytes = 7
/project/src/tests/tests.c:33:TestLeds_OnOff:PASS

-----------------------
1 Tests 0 Failures 0 Ignored
OK
INFO  McuUnity.c:89 *** PASSED ***
*STOP*0
=============================================
Target RTT wildcard exit value 0.
```

Github actions

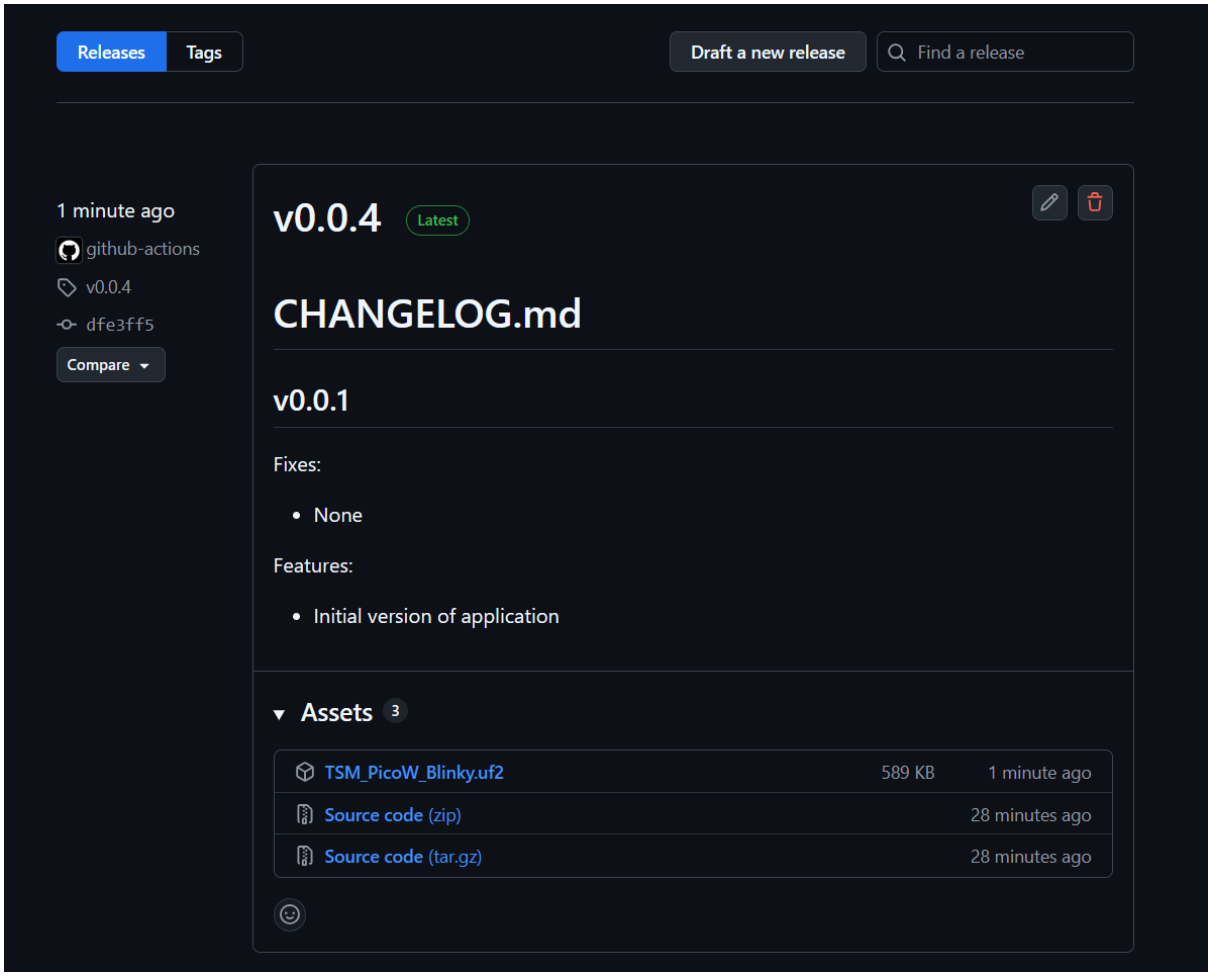
Deploy.yml is doing the following:

1. Defines possible triggers to come either from a push or manually
2. Checks the project name to match CMakeLists.txt
3. Defines the steps of a job run, which are: Checkout repo, copy dockerifle, build docker image, create container, copy uf2 to the runners workspace, exports the APP_NAME to the github environment so the next step can reference it, publish a GitHub Release that contains the .uf2 and CHANGELOG.md as release notes

Running the job manually:



Adding manually a new release note, tag, and push it. This triggers the action runner automatically and creates a new release on github:



Discussion

1. You might notice that the tests run differently on your host compared to the tests with the docker container (address of argument passed to the tests). What could be the reason for this? What would be the solution?

-> The reason is that the host and Docker container use different toolchains. Your Windows host uses your locally installed ARM GCC, while Docker uses the xPack GCC 13.2.1 downloaded during the image build. Even with the same GCC version, differences in the linker, build flags, or host OS can produce a binary with a slightly different memory layout.

The solution is to always use the same ELF that was built in Docker for running the tests

2. What are your key findings running this CI/CD integration as outlined in this lab for an embedded target? What could be improved or what is needed to solve the challenges?

-> the automated tests are a huge advantage of this workflow method. Doing it for embedded systems is a bit harder, but with the correct setup of the chain consisting of the tunnel that bridges between docker and target it works fine. There may be problems with the network because of firewalls etc.

3. Both GitHub and GitLab provide a CI/CD integration. While offering both similar things, they are different. What is your experience or opinion about this?

-> Both use YAML pipeline definitions and Docker-based runners, so the concepts transfer well.

GitLab's pipeline syntax is more powerful for complex workflows — stages, needs, rules, and only/except give fine-grained control over job dependencies and trigger conditions. It also has a built-in container registry and can be fully self-hosted, which is a significant advantage for embedded teams who need a runner physically attached to hardware. You can run the entire GitLab instance on a local server, keep the probe on USB, and never deal with tunnel issues at all.

GitHub Actions has a much larger ecosystem of community actions and tighter integration with the GitHub release/package system (as seen in this lab with softprops/action-gh-release). The workflow_dispatch trigger and the release pipeline feel natural for open-source.

Repository

My code can be found on: https://github.com/cyrillaebische/TSM_SwEng_Embedded-CICD/settings